ML Optimisation Progress Documentation

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Model Performance on Training Data

Model Testing 1 (21/2/26)

```
--- Core performance metrics ---

[Test 1] Accuracy: 79.34%

[Test 2] Weighted F1-score: 79.27%

[Test 3] Classification Report:
              precision    recall  f1-score   support

     Comment       1.00      0.94      0.97        17
        Like       0.60      0.61      0.61        49
        Play       0.90      0.94      0.92        50
      Search       1.00      1.00      1.00        47
   Subscribe       0.60      0.58      0.59        50

    accuracy                           0.79       213
   macro avg       0.82      0.81      0.82       213
weighted avg       0.79      0.79      0.79       213

[Test 4] Confusion Matrix:
[[16  0  1  0  0]
 [ 0 30  2  0 17]
 [ 0  1 47  0  2]
 [ 0  0  0 47  0]
 [ 0 19  2  0 29]]

[Test 5] ROC-AUC Score:
ROC-AUC Score (multi-class, OVR): 0.9557

[Test 6] Per-class accuracy:
Comment: 94.12%
Like: 61.22%
Play: 94.00%
Search: 100.00%
Subscribe: 58.00%
```

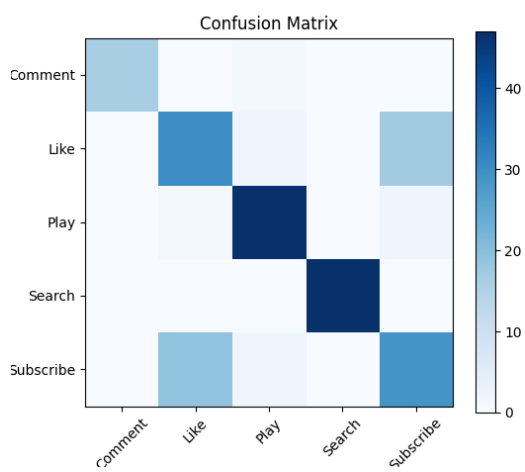
```
[Test 8] Permutation Importance:
duration              2.638498e-01
std_iat               5.164319e-02
avg_outbound_size     1.408451e-02
avg_iat               7.042254e-03
pk_count              2.816901e-03
avg_inbound_size      1.110223e-17
total_bytes          -1.408451e-03
outbound_ratio       -4.694836e-03
std_pkt_size         -5.164319e-03
avg_pkt_size         -5.164319e-03
max_pkt_size         -7.981221e-03
dtype: float64

[Test 9] Feature ablation results:
Baseline accuracy: 79.34%
Without pk_count: 79.34%
Without total_bytes: 75.12%
Without avg_pkt_size: 78.40%
Without std_pkt_size: 76.53%
Without max_pkt_size: 78.40%
Without avg_iat: 78.87%
Without std_iat: 77.46%
Without duration: 77.93%
```

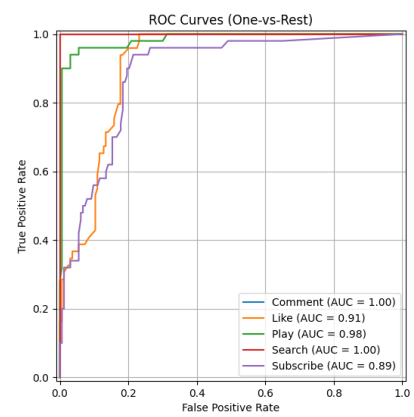
```
[Test 11] Class distribution:
Total samples: 645
action
Like         150
Play         150
Subscribe    150
Search       143
Comment       52
Name: count, dtype: int64

[Test 12]Prediction confidence (first 10 rows):
Sample 0: 0.910
Sample 1: 0.605
Sample 2: 0.970
Sample 3: 0.940
Sample 4: 0.670
Sample 5: 1.000
Sample 6: 0.495
Sample 7: 0.600
Sample 8: 0.835
Sample 9: 1.000

[Test 13] Noise sensitivity test:
Noise 0.5% -> Accuracy: 81.22%
Noise 1.0% -> Accuracy: 77.46%
Noise 5.0% -> Accuracy: 78.87%
```

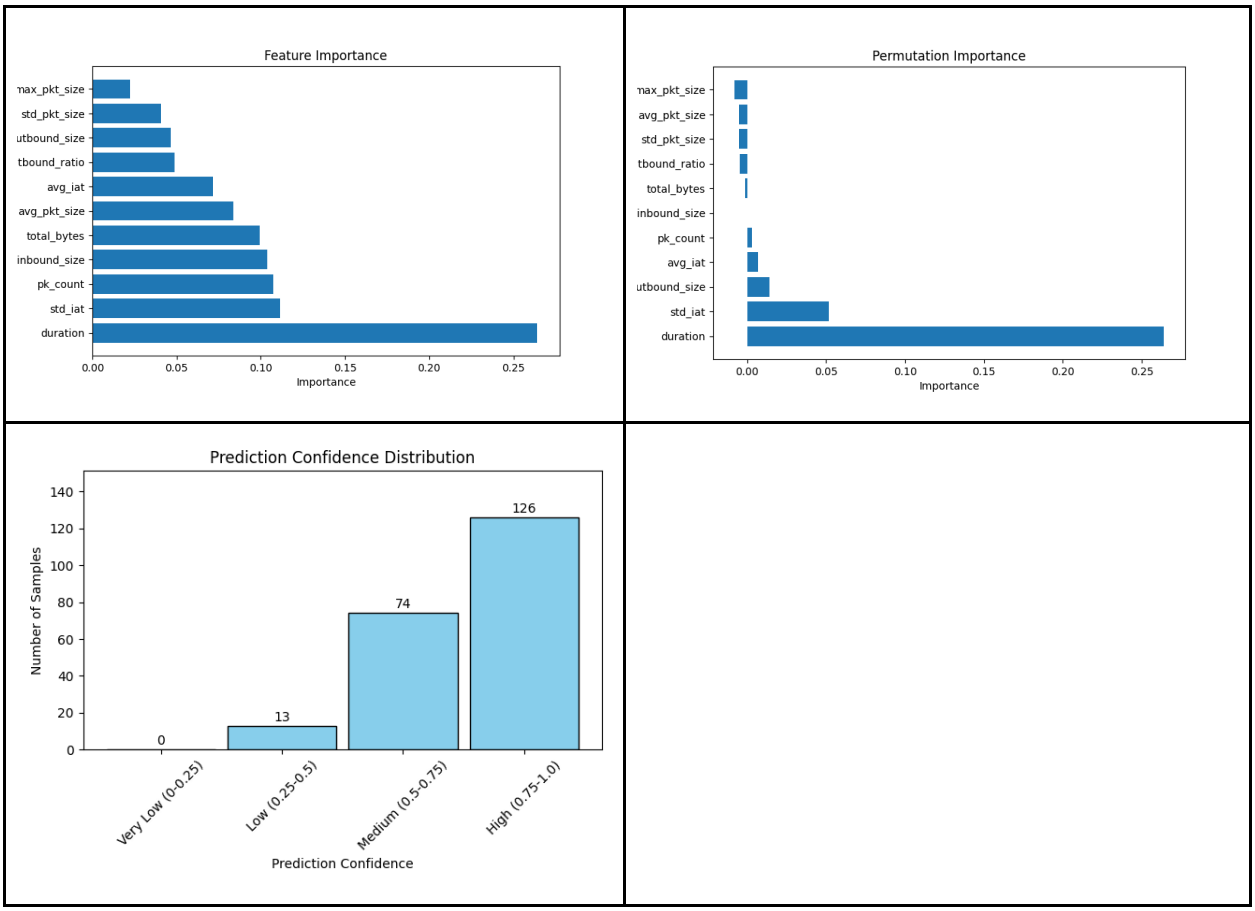
```
[Test 14] Stability test:

Stability scores: [0.8169014084507042, 0.82629
10798122066, 0.8262910798122066, 0.81690140845
07042, 0.784037558685446]
Mean stability: 0.8140845070422535
```

```
--- Model robustness & reliability ---

[Test 10] Overfitting check:

Training Accuracy: 100.00%
Testing Accuracy:  79.34%
```

```
--- Error & unseen label analysis ---

[Test 15] Error analysis:

Misclassified samples:
     pk_count  total_bytes  ...       true       pred
380      2079      2275493  ...  Subscribe       Like
86       4981      5571646  ...       Like  Subscribe
429      2322      2512597  ...  Subscribe       Like
372      1839      1949806  ...  Subscribe       Like
129      3728      4117888  ...       Like  Subscribe
416      2576      2786165  ...  Subscribe       Like
395      5168      5824475  ...  Subscribe       Like
432      6049      6915035  ...  Subscribe       Like
83       5571      6361361  ...       Like  Subscribe
366      7274      8205666  ...  Subscribe       Like
402      2084      2161600  ...  Subscribe       Like
220      1572      6319154  ...       Play  Subscribe
373      3021      3204798  ...  Subscribe       Like
61       4129      4476640  ...       Like       Play
87       2644      2916359  ...       Like  Subscribe
8         872      1900734  ...       Like  Subscribe
149      1465      1552587  ...       Like  Subscribe
110      2866      3123472  ...       Like  Subscribe
423      6349      7213319  ...  Subscribe       Like
70       1727      1775297  ...       Like  Subscribe

[20 rows x 13 columns]
Total misclassified: 44

[Test 16] Testing model handling of unseen labels:

Unknown label handling test passed.
Model correctly raises error for unseen class:
Error: y contains previously unseen labels: 'new_action'

[test 17] General Sanity check (first 5 rows of features):
     pk_count  total_bytes  ...  avg_outbound_size  avg_inbound_size
485      7342      8039065  ...         235.863911       1229.147717
449      4782      5405458  ...         188.973638       1257.520057
460     10287     11589139  ...         158.495955       1258.782234
488      4328      4424436  ...         332.895161       1165.391183
345       741      1907284  ...         634.190476       3573.554192

[5 rows x 11 columns]
```







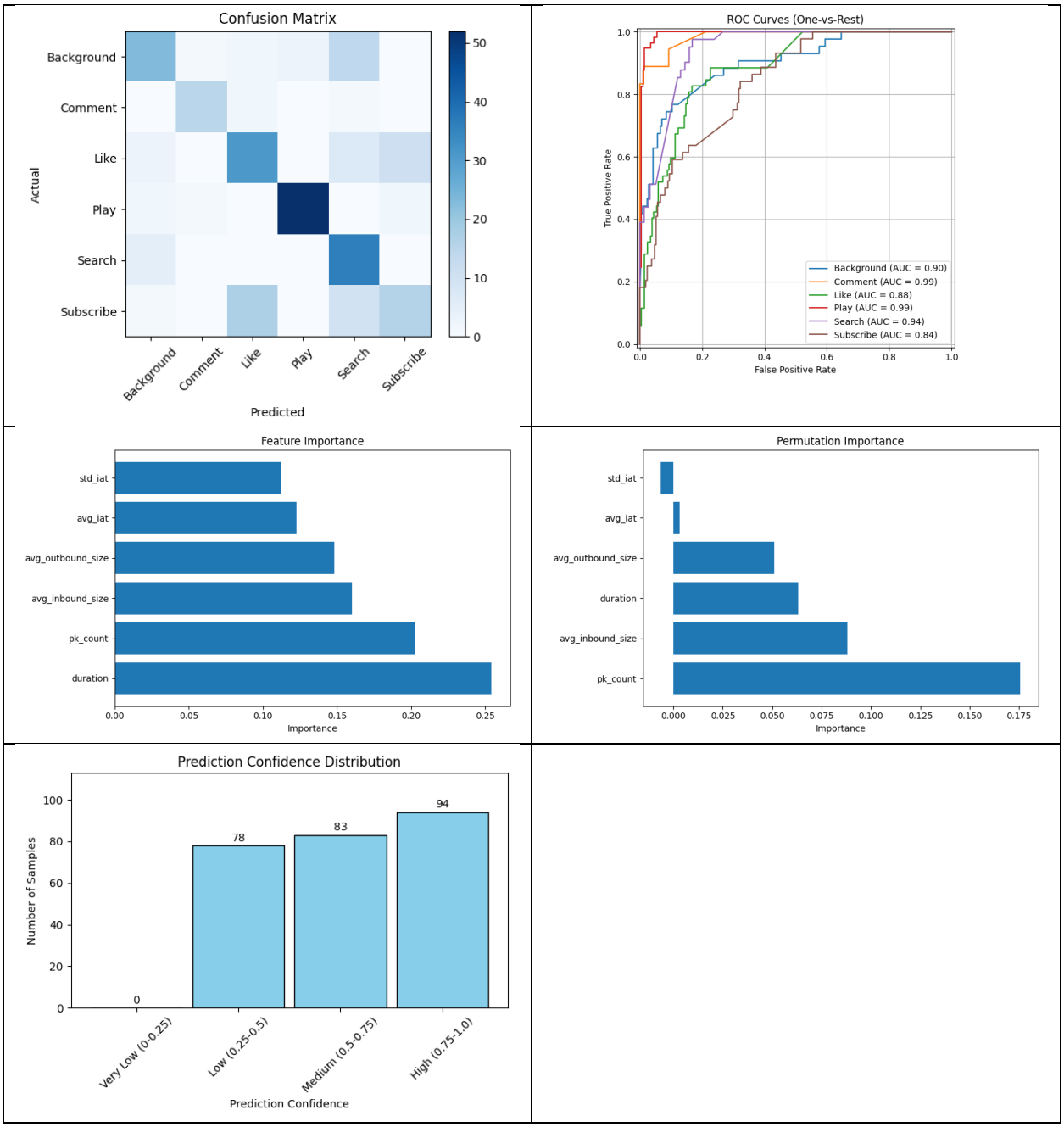
Model Testing 2 (9/3/26)

```
--- Running Model Evaluation ---

--- Core performance metrics ---

[Test 1] Accuracy: 67.84%

[Test 2] Weighted F1-score: 67.22%

[Test 3] Classification Report:
              precision    recall  f1-score   support

  Background       0.68      0.53      0.60        43
     Comment       0.88      0.83      0.86        18
        Like       0.60      0.60      0.60        52
        Play       0.95      0.91      0.93        57
      Search       0.53      0.88      0.66        41
   Subscribe       0.55      0.36      0.44        44

    accuracy                           0.68       255
   macro avg       0.70      0.69      0.68       255
weighted avg       0.69      0.68      0.67       255

[Test 4] Confusion Matrix:
[[23  1  2  3 14  0]
 [ 0 15  2  0  1  0]
 [ 3  0 31  0  7 11]
 [ 2  1  0 52  0  2]
 [ 5  0  0  0 36  0]
 [ 1  0 17  0 10 16]]

[Test 5] ROC-AUC Score:
ROC-AUC Score (multi-class, OVR): 0.9235

[Test 6] Per-class accuracy:
Background: 53.49%
Comment: 83.33%
Like: 59.62%
Play: 91.23%
Search: 87.80%
Subscribe: 36.36%


--- Feature analysis ---

[Test 7] Top Feature Importances:
duration             0.254155
pk_count             0.202493
avg_inbound_size     0.160064
avg_outbound_size    0.148323
avg_iat              0.122425
std_iat              0.112540
dtype: float64
```

```
[Test 8] Permutation Importance:
pk_count             0.175686
avg_inbound_size     0.088235
duration             0.063137
avg_outbound_size    0.050980
avg_iat              0.003137
std_iat             -0.006275
dtype: float64

[Test 9] Feature ablation results:
Baseline accuracy: 67.84%
Without duration: 67.06%
Without std_iat: 69.02%
Without avg_iat: 67.06%
Without pk_count: 65.88%
Without avg_inbound_size: 67.84%
Without avg_outbound_size: 67.84%


--- Model robustness & reliability ---

[Test 10] Overfitting check:

Training Accuracy: 77.38%
Testing Accuracy:  67.84%

[Test 11] Class distribution:
Total samples: 790
action
Like          150
Play          150
Subscribe     150
Background    150
Search        138
Comment        52
Name: count, dtype: int64

[Test 12]Prediction confidence (first 10 rows):
Sample 0: 0.676
Sample 1: 0.667
Sample 2: 0.530
Sample 3: 0.779
Sample 4: 0.450
Sample 5: 0.475
Sample 6: 0.688
Sample 7: 0.288
Sample 8: 0.450
Sample 9: 0.283

[Test 13] Noise sensitivity test:
Noise 0.5% -> Accuracy: 68.24%
Noise 1.0% -> Accuracy: 68.24%
Noise 5.0% -> Accuracy: 68.63%
```

```
[Test 14] Stability test:

Stability scores: [0.7241379310344828, 0.7164750957854407, 0.7279693486590039, 0.7394636015325671, 0.7126436781609196]
Mean stability: 0.7241379310344828


--- Error & unseen label analysis ---

[Test 15] Error analysis:

Misclassified samples:
      duration   std_iat   avg_iat  ...  avg_outbound_size  true        pred
202   1.402640  0.122197  0.035066  ...         128.375000  Like  Background
265   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
489   2.632032  0.138046  0.039879  ...         775.848485  Like   Subscribe
630   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
764   4.100947  0.188961  0.048246  ...         705.047059  Like   Subscribe
860   0.000000  0.000000  0.000000  ...          66.000000  Like      Search
892   0.013041  0.004767  0.002608  ...         110.400000  Like      Search
1154  1.495321  0.583267  0.249220  ...         145.000000  Like  Background
1338  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
1432  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
1544  2.278423  0.126649  0.052987  ...        2656.372093  Like   Subscribe
1594  2.332916  0.126422  0.044017  ...        1927.886792  Like   Subscribe
1653  2.297378  0.103865  0.028717  ...         679.475000  Like   Subscribe
1743  2.985846  0.122321  0.040902  ...         782.232877  Like   Subscribe
1934  1.617230  0.129754  0.041467  ...         118.179487  Like  Background
2114  2.841430  0.099818  0.033040  ...         710.290698  Like   Subscribe
2221  3.127080  0.102260  0.039089  ...         720.462500  Like   Subscribe
2267  0.000000  0.000000  0.000000  ...          66.000000  Like      Search
2322  3.250708  0.186127  0.056047  ...        2236.655172  Like   Subscribe
2354  2.341362  0.236726  0.055747  ...        2138.880952  Like   Subscribe

[20 rows x 8 columns]
Total samples: 255
Total misclassified: 82
```

```
[Test 16] Testing model handling of unseen labels:

Unknown label handling test passed.
Model correctly raises error for unseen class:
Error: y contains previously unseen labels: 'new_action'

[test 17] General Sanity check (first 5 rows of features):
     duration   std_iat   avg_iat  pk_count  avg_inbound_size  avg_outbound_size
94   1.918529  0.116340  0.042634        45               0.0        2373.488889
163  1.970391  0.118672  0.042835        46               0.0        2303.956522
202  1.402640  0.122197  0.035066        40               0.0         128.375000
239  2.449518  0.116469  0.035500        69               0.0         682.898551
265  0.000000  0.000000  0.000000         1               0.0          66.000000

--- Testing complete ---
```



Model Testing 3 (13/3/26)

Added features

1. Max packet size
2. Packet burst standard deviation
3. Packet count ratio

Adjusted all constraints on how the system recognises action flows

```
[Test 1] Accuracy: 45.29%

[Test 2] Weighted F1-score: 46.42%

[Test 3] Classification Report:
              precision    recall  f1-score   support

  Background       0.32      0.50      0.39       204
     Comment       0.55      0.53      0.54       571
        Like       0.28      0.54      0.37       243
        Play       0.78      0.34      0.48       882
      Search       0.39      0.85      0.53        27
   Subscribe       0.31      0.51      0.39       228

    accuracy                           0.45      2155
   macro avg       0.44      0.54      0.45      2155
weighted avg       0.56      0.45      0.46      2155

[Test 4] Confusion Matrix:
[[102  39   1  61   1   0]
 [  0 300 136  26  16  93]
 [  1  21 131   0   6  84]
 [217 166 116 304   4  75]
 [  0   1   0   1  23   2]
 [  0  14  89   0   9 116]]

[Test 5] ROC-AUC Score:
ROC-AUC Score (multi-class, OVR): 0.8566

[Test 6] Per-class accuracy:
Background: 50.00%
Comment: 52.54%
Like: 53.91%
Play: 34.47%
Search: 85.19%
Subscribe: 50.88%
```

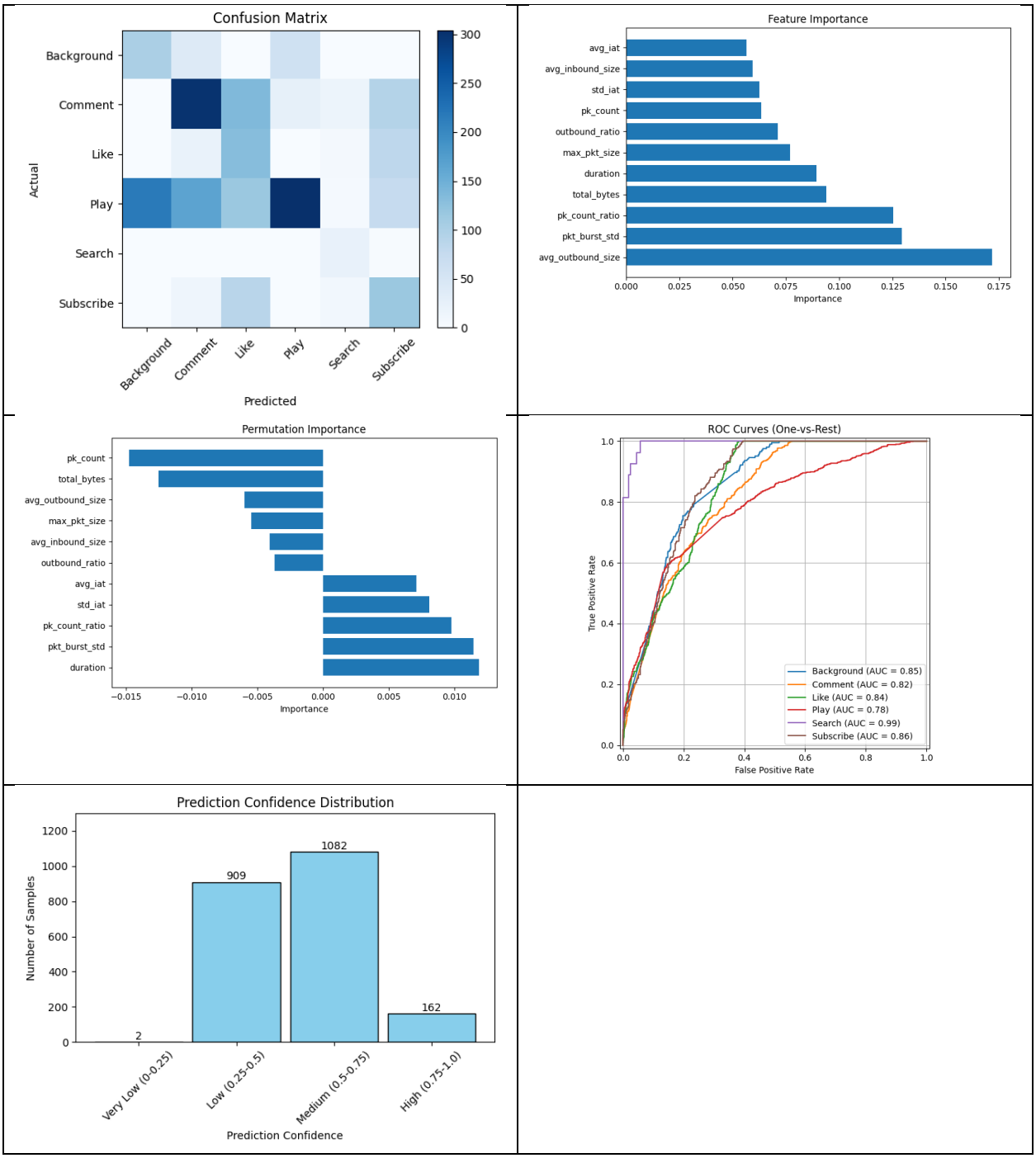
```
[Test 7] Top Feature Importances:
avg_outbound_size    0.171824
pkt_burst_std        0.129585
pk_count_ratio       0.125298
total_bytes          0.094104
duration             0.089256
max_pkt_size         0.076989
outbound_ratio       0.071257
pk_count             0.063393
std_iat              0.062527
avg_inbound_size     0.059279
avg_iat              0.056488
dtype: float64

[Test 8] Permutation Importance:
duration             0.011879
pkt_burst_std        0.011415
pk_count_ratio       0.009745
std_iat              0.008074
avg_iat              0.007100
outbound_ratio      -0.003712
avg_inbound_size    -0.004084
max_pkt_size        -0.005476
avg_outbound_size   -0.005986
total_bytes         -0.012529
pk_count            -0.014756
dtype: float64

[Test 9] Feature ablation results:
Baseline accuracy: 45.29%
Without duration: 45.29%
Without std_iat: 45.52%
Without avg_iat: 45.10%
Without pk_count: 45.20%
Without max_pkt_size: 45.06%
Without pkt_burst_std: 45.20%
Without pk_count_ratio: 44.97%
Without avg_inbound_size: 45.71%
Without avg_outbound_size: 45.48%
Without total_bytes: 44.97%
Without outbound_ratio: 45.71%
```

```
[Test 10] Overfitting check:

Training Accuracy: 58.35%
Testing Accuracy:  45.29%

[Test 11] Class distribution:
Total samples: 6618
action
Play          2892
Comment       1491
Subscribe      780
Like           769
Background     600
Search          86
Name: count, dtype: int64

[Test 12]Prediction confidence (first 10 rows):
Sample 0: 0.483
Sample 1: 0.577
Sample 2: 0.455
Sample 3: 0.577
Sample 4: 0.494
Sample 5: 0.375
Sample 6: 0.407
Sample 7: 0.387
Sample 8: 0.512
Sample 9: 0.498

[Test 13] Noise sensitivity test:
Noise 0.5% -> Accuracy: 45.15%
Noise 1.0% -> Accuracy: 45.15%
Noise 5.0% -> Accuracy: 44.32%
```

```
[Test 14] Stability test:

Stability scores: [0.5842490842490843, 0.5938644688644689, 0.603021978021978, 0.5856227106227107, 0.5975274725274725]
Mean stability: 0.5928571428571429
```

```
[Test 15] Error analysis:

Misclassified samples:
     duration   std_iat   avg_iat  pk_count  ...  total_bytes  outbound_ratio  true       pred
117  0.083948  0.022603  0.016790         5  ...         1043             1.0  Like  Subscribe
120  0.105001  0.012162  0.007580        14  ...         4271             1.0  Like     Search
127  0.012967  0.004336  0.002161         6  ...          566             1.0  Like  Subscribe
128  0.901520  0.287676  0.225380         4  ...         5168             1.0  Like  Subscribe
129  0.022035  0.007775  0.003673         6  ...          676             1.0  Like  Subscribe
181  0.030293  0.008610  0.006059         5  ...         2356             1.0  Like    Comment
183  1.807620  0.380704  0.180762        10  ...         3304             1.0  Like  Subscribe
187  2.102936  0.577448  0.420587         5  ...         6460             1.0  Like  Subscribe
188  0.038458  0.014358  0.006410         6  ...          797             1.0  Like  Subscribe
190  0.019089  0.004486  0.003818         5  ...          602             1.0  Like  Subscribe
290  0.077196  0.012403  0.008577         9  ...         3672             1.0  Like    Comment
308  2.101237  0.502203  0.420247         5  ...         6460             1.0  Like  Subscribe
309  1.253631  0.068993  0.025073        50  ...         5815             1.0  Like  Subscribe
311  0.112448  0.022455  0.016064         7  ...         3642             1.0  Like    Comment
317  0.021028  0.010138  0.005257         4  ...          414             1.0  Like  Subscribe
325  0.024692  0.009941  0.004938         5  ...          534             1.0  Like  Subscribe
502  3.091692  0.359157  0.154585        20  ...         2929             1.0  Like  Subscribe
503  2.026491  0.104948  0.025331        80  ...        10183             1.0  Like  Subscribe
508  0.049667  0.025707  0.016556         3  ...         1430             1.0  Like    Comment
510  0.086923  0.012499  0.007902        11  ...         5870             1.0  Like    Comment

[20 rows x 13 columns]
Total samples: 2155
Total misclassified: 1179
```

```
[Test 16] Testing model handling of unseen labels:

Unknown label handling test passed.
Model correctly raises error for unseen class:
Error: y contains previously unseen labels: 'new_action'

[test 17] General Sanity check (first 5 rows of features):
     duration   std_iat   avg_iat  pk_count  ...  avg_inbound_size  avg_outbound_size  total_bytes  outbound_ratio
107  1.749035  0.516872  0.159003        11  ...               0.0         127.818182         1406             1.0
113  2.102404  0.502391  0.420481         5  ...               0.0        1292.000000         6460             1.0
134  1.033926  0.072411  0.010883        52  ...               0.0         114.500000         5954             1.0
116  2.101976  0.502487  0.420395         5  ...               0.0        1292.000000         6460             1.0
117  0.083948  0.022603  0.016790         5  ...               0.0         208.600000         1043             1.0

[5 rows x 11 columns]
```



Model Performance on User Uploaded PCAP (Changes + Tests)

Automated testing results

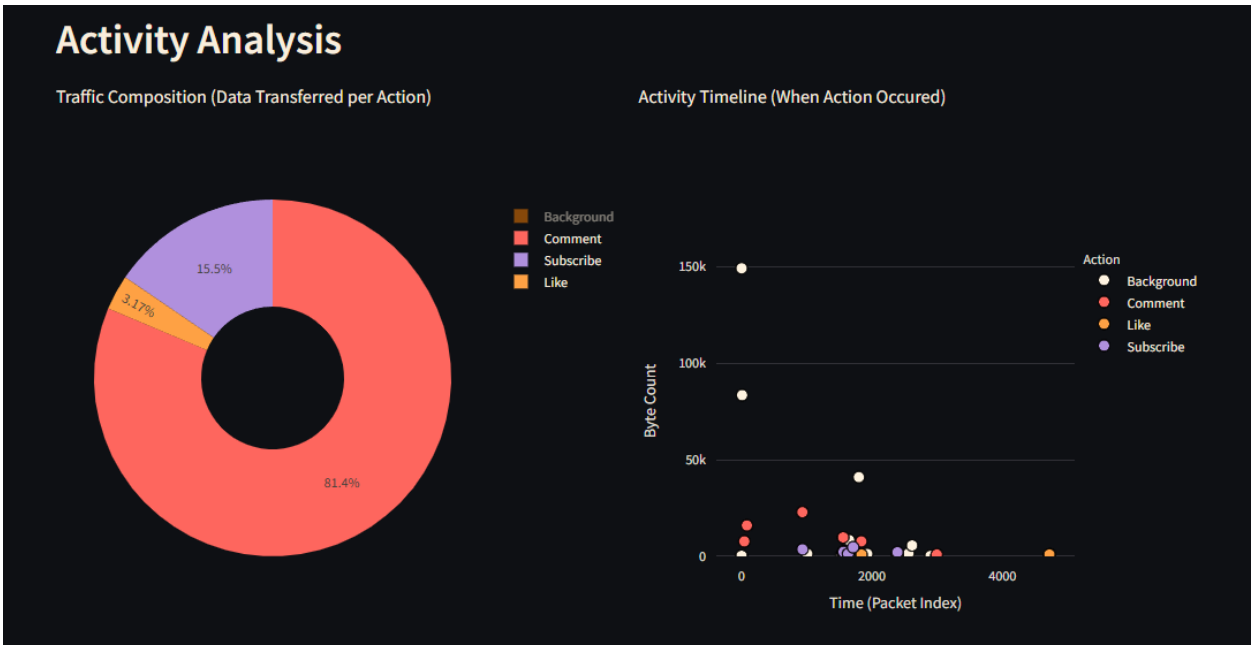
Modified: 12/03/2026

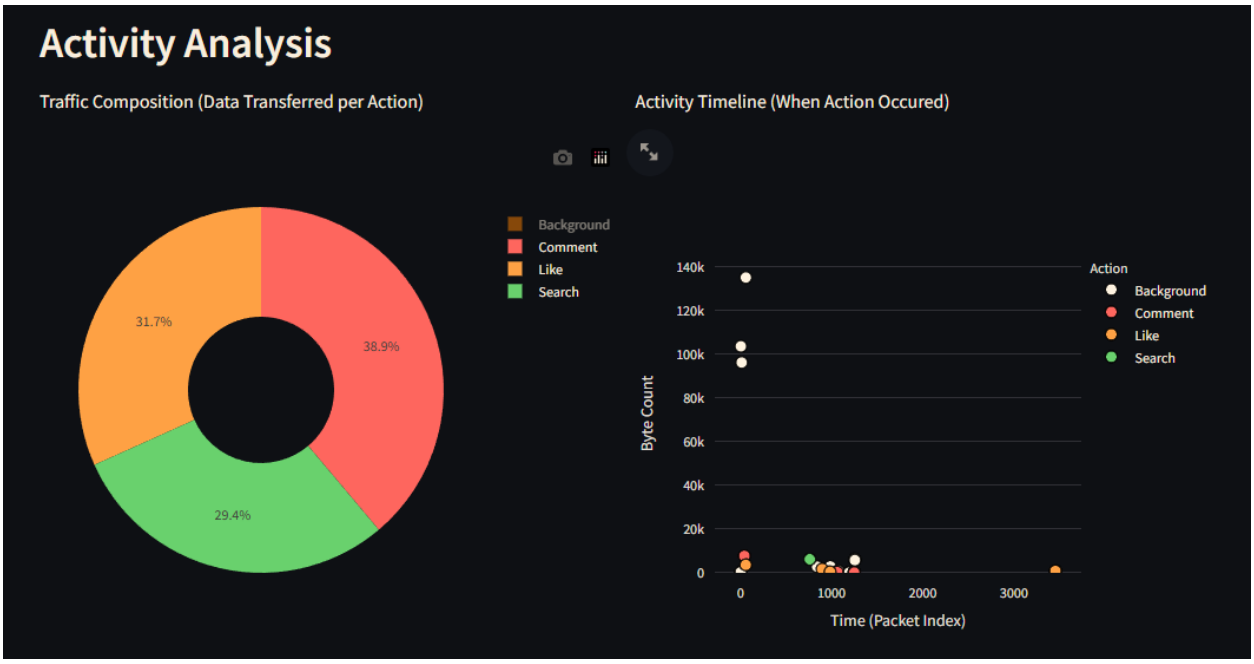
```
==================================================================
SUMMARY
==================================================================
Action        Pass  Fail   Pass Rate
------------------------------------------------
Comment        113    37         75%
Like            94    56         63%
Play             1   149          1%
Search          32   118         21%
Subscribe       78    72         52%
------------------------------------------------
TOTAL          318   432         42%
==================================================================
```

Results

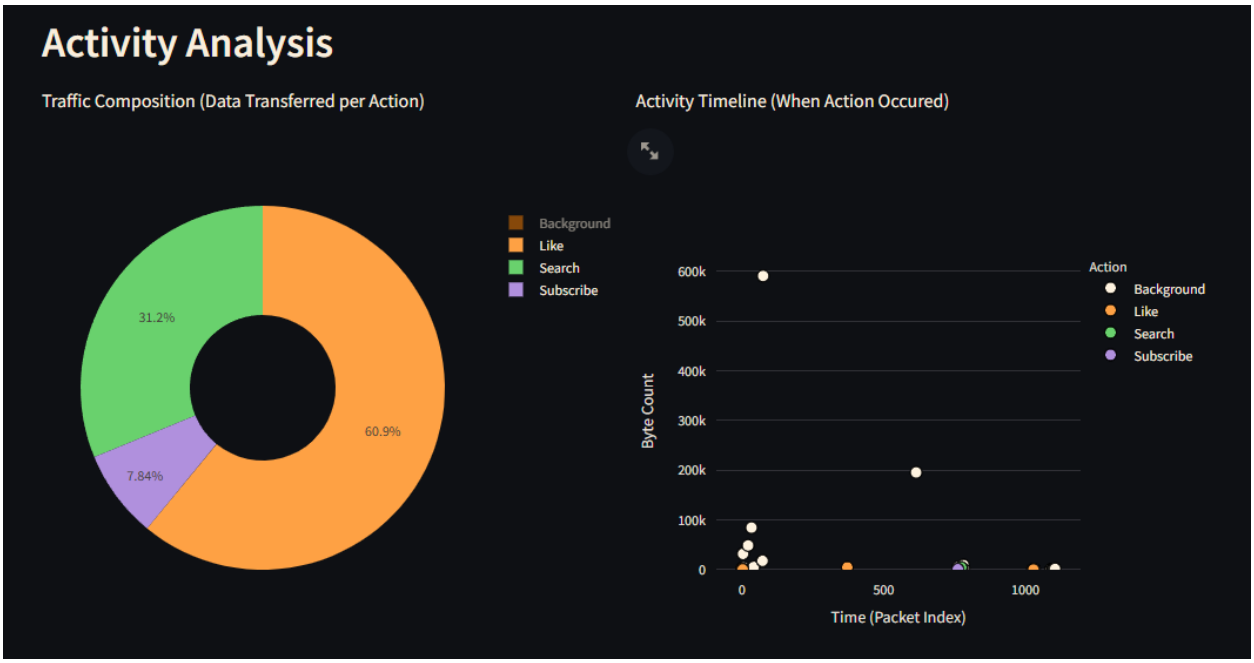
For all results in this section, the PCAP file from the dataset with the numbers 0 and 10 havebeen uploaded onto the dashboard and the background action has been hidden from the pie chart for better visibility (modified: 12/03/2026).

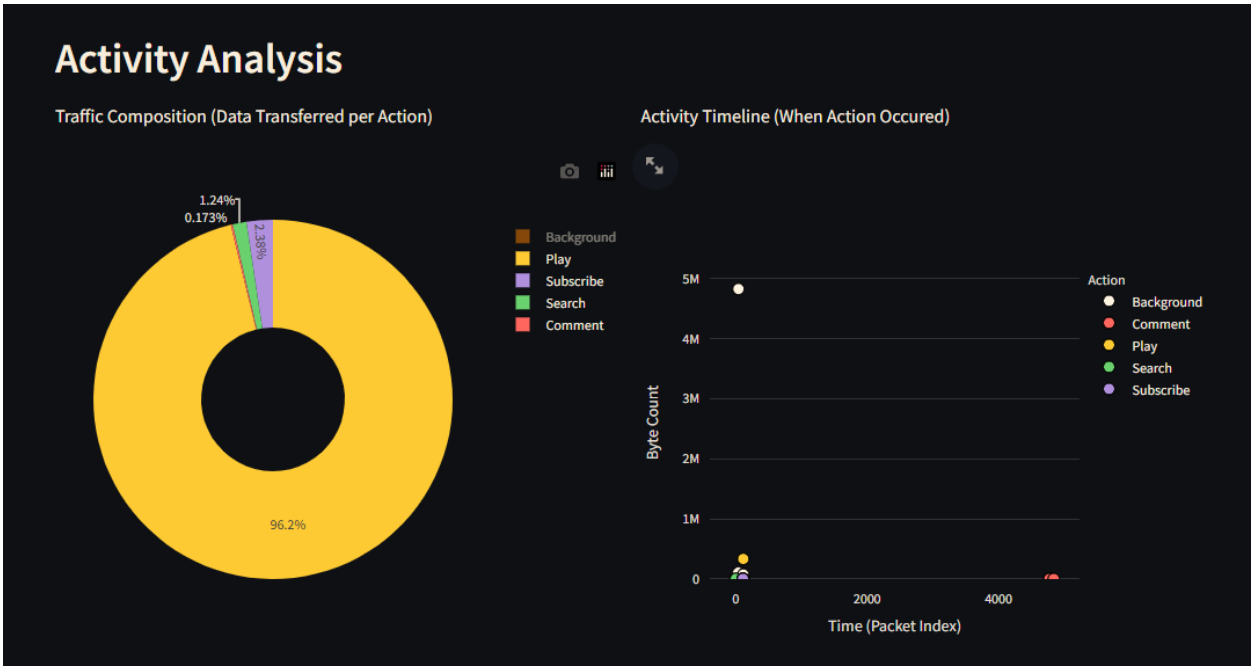
Comment:


Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
Background
Comment
Subscribe
Like
15.5%
3.17%
81.4%
Byte Count
150k
100k
50k
0
0
2000
4000
Time (Packet Index)
Action
Background
Comment
Like
Subscribe

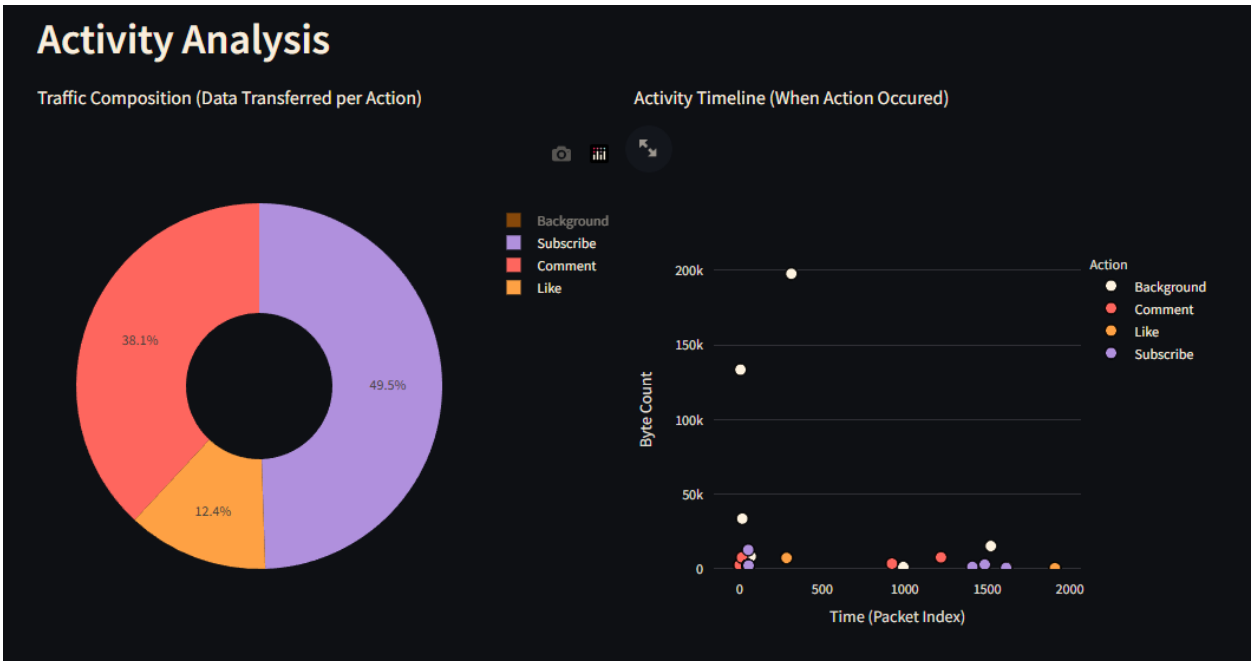

Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
Background
Comment
Like
Search
31.7%
38.9%
29.4%
Byte Count
140k
120k
100k
80k
60k
40k
20k
0
0
1000
2000
3000
Time (Packet Index)
Action
Background
Comment
Like
Search

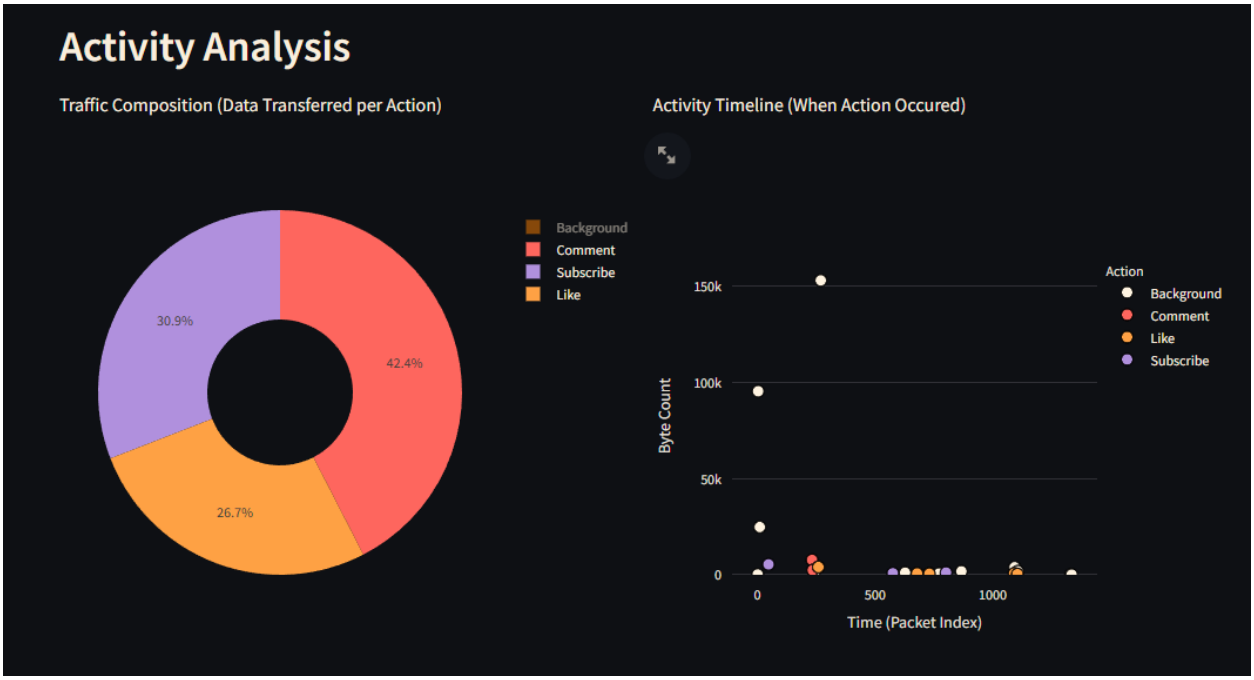
Like:


Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
Background
Like
Search
Subscribe
31.2%
60.9%
7.84%
Byte Count
600k
500k
400k
300k
200k
100k
0
0
500
1000
Time (Packet Index)
Action
Background
Like
Search
Subscribe


Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
1.24%
0.173%
2.38%
96.2%
Background
Play
Subscribe
Search
Comment
Byte Count
5M
4M
3M
2M
1M
0
0
2000
4000
Time (Packet Index)
Action
Background
Comment
Play
Search
Subscribe

Play:

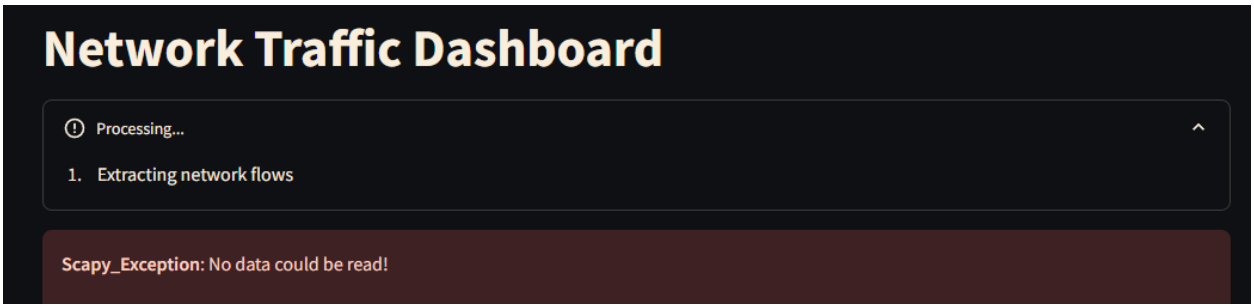


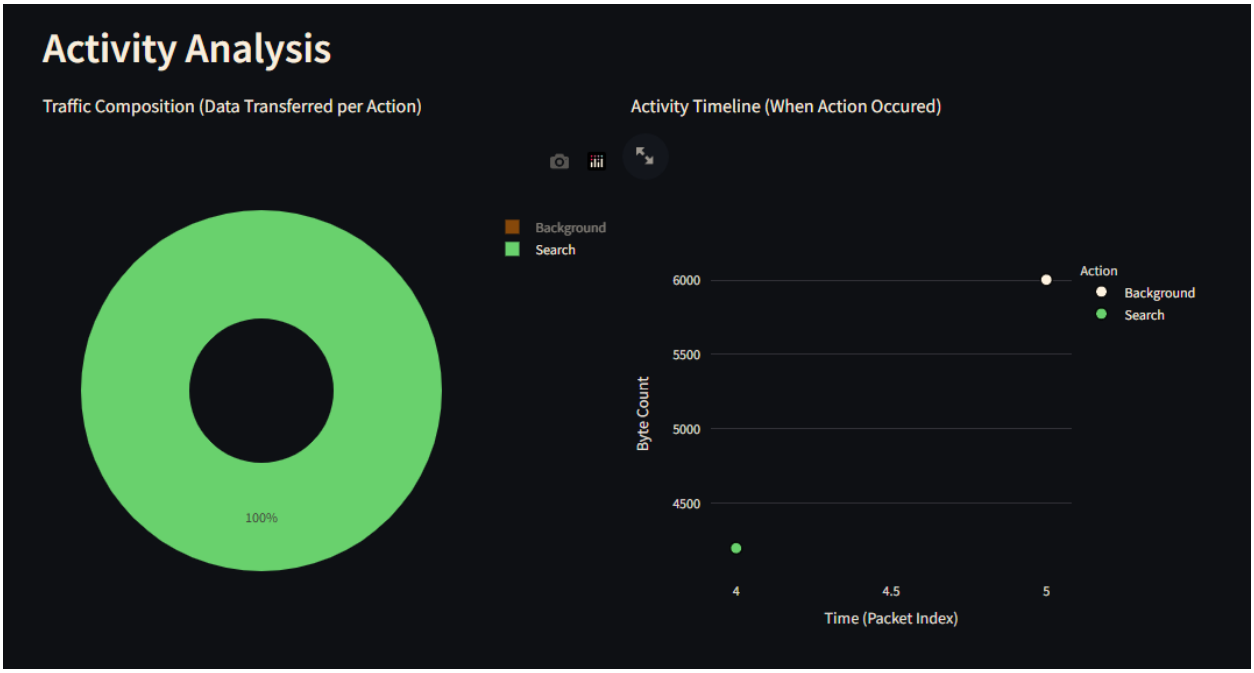


Search:

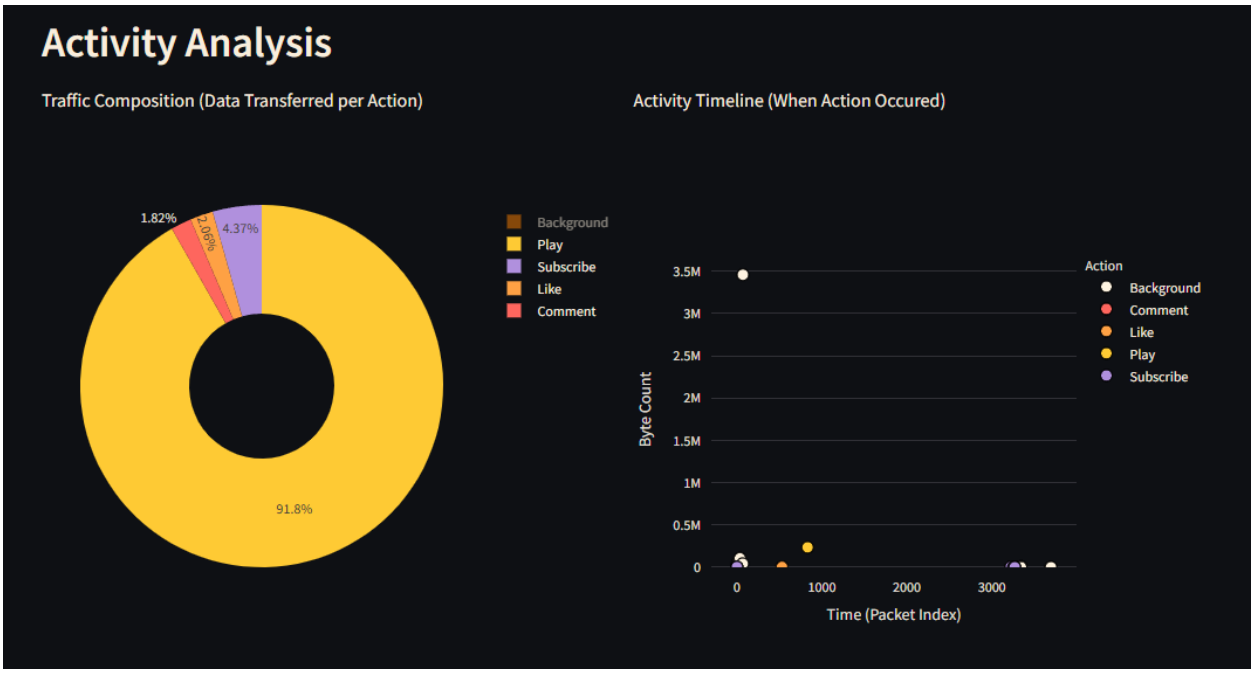
- very little flows for each Search PCAP in datasets

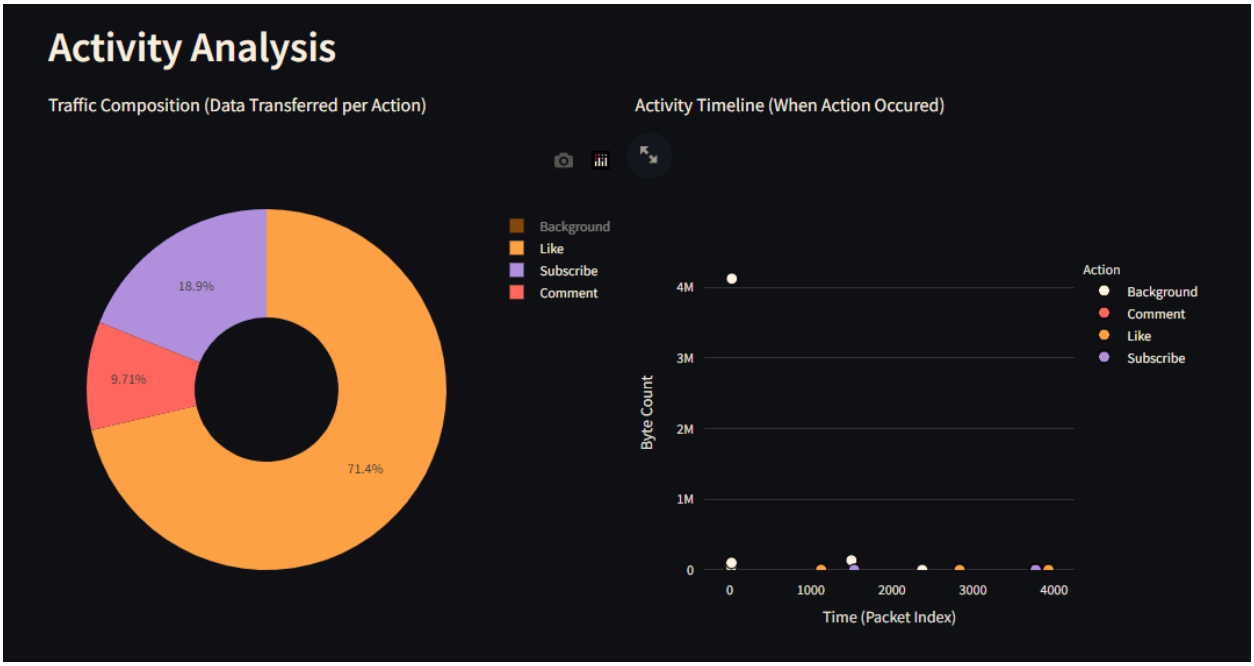
- exception error for some PCAPs





Subscribe:


Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
1.82%
2.06%
4.37%
91.8%
Background
Play
Subscribe
Like
Comment
Byte Count
3.5M
3M
2.5M
2M
1.5M
1M
0.5M
0
0
1000
2000
3000
Time (Packet Index)
Action
Background
Comment
Like
Play
Subscribe


Activity Analysis
Traffic Composition (Data Transferred per Action)
Activity Timeline (When Action Occured)
18.9%
9.71%
71.4%
Background
Like
Subscribe
Comment
Byte Count
4M
3M
2M
1M
0
0
1000
2000
3000
4000
Time (Packet Index)
Action
Background
Comment
Like
Subscribe